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Session : Technical Paper Presentation

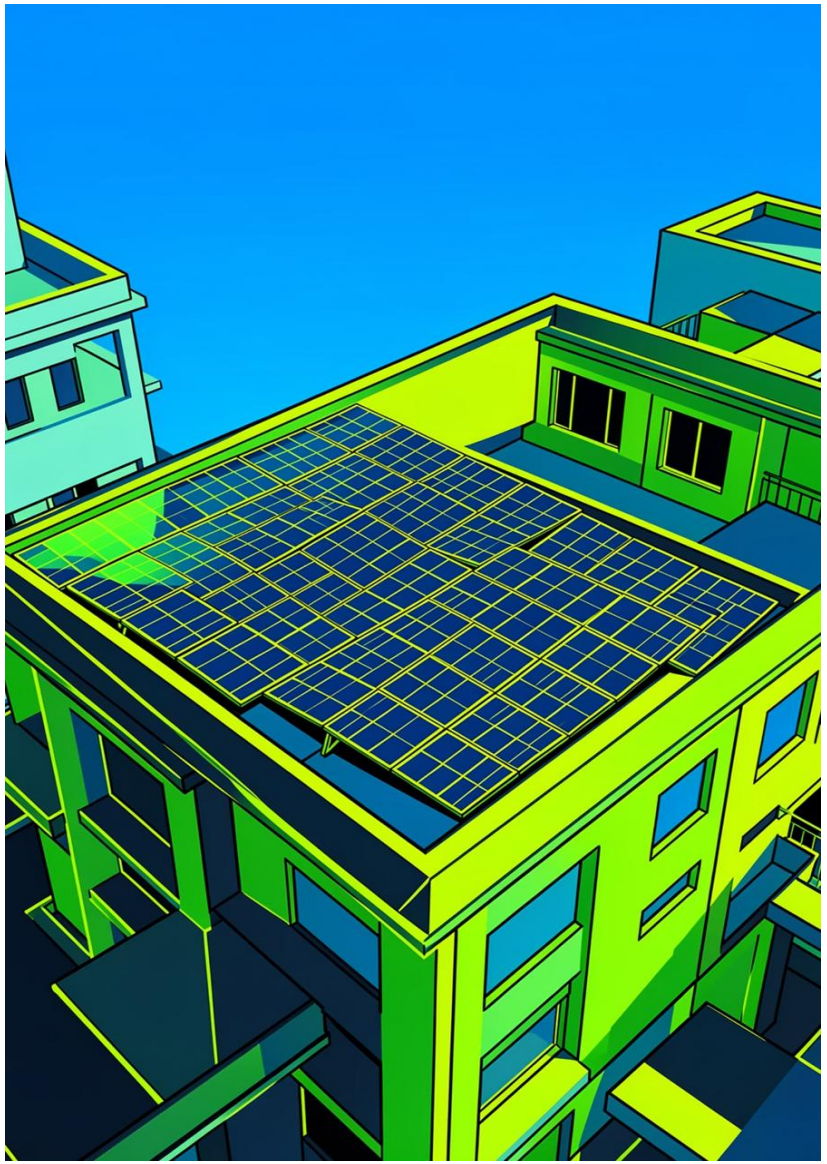
Accelerating India's Solar Mission: A Tech Enabled Customer Engagement Model Under PM Surya Ghar

Presented By

RANJEET KUMAR,

MANAGER, BSES Rajdhani Power Limited

INTRODUCTION



Accelerating India's Solar Mission: A Tech Enabled Customer Engagement Model Under PM Surya Ghar"

Renewable Energy | *Roof Top Solar* | *RTSE Platform*

1. Address low awareness, complex financing, and fragmented ecosystem for rooftop solar adoption.
2. RTSE platform streamlines engagement, simplifies applications, offers financial guidance, and connects with certified vendors
3. Increased solar adoption, reduced household energy costs, and positive environmental impact
4. Empower households for energy independence, reduced emissions, and a sustainable future

CONTEXT & BACKGROUND

The PM Surya Ghar Yojana is more than just a scheme; it's a strategic move to address critical national priorities and empower citizens.

PM Surya Ghar Scheme

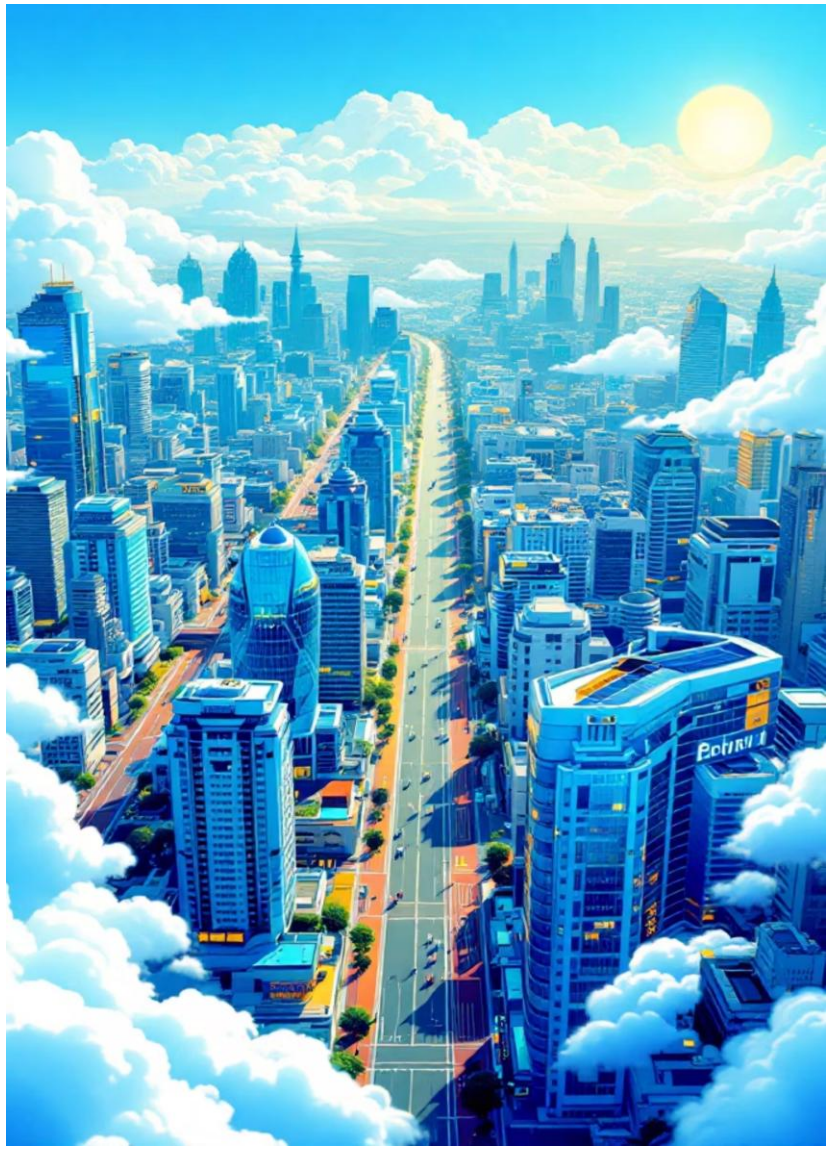
- Launched on **February 29,2024** By PM Modi
- **10 Million** Households Target
- **Up to 60% Subsidy** Coverage by Government
- **75k ..Cr Total Investment** backing this revolution
- **300 Units of Free Electricity** per month for qualifying families
- **COP26 Target-** Help India to reach **500 GW Non-Fossil Fuel Capacity by 2030 & Net Zer0 by 2070**

Delhi's Opportunity

High solar resource potential but low rooftop adoption due to finance, policy, and awareness gaps

RTSE Platform

Geospatial tool combining drone/satellite imagery, DSM, and 3D point clouds for rooftop-level assessments.



RELEVANCE

The PM Surya Ghar Yojana is more than just a scheme; it's a strategic move to address critical national priorities and empower citizens

Energy Independence & Security

Reducing India's significant fossil fuel import dependency (currently **over 80%**) and enhancing national energy security by harnessing its vast solar potential of **748 GW**.

Climate Action & Sustainability

Mitigating climate change impacts and curbing pollution, aiming for a reduction of **45% in emission intensity by 2030** and an estimated **720 million tonnes of CO2 reduction** over the scheme's lifetime

Household Empowerment & Savings

Making clean energy accessible and affordable for middle-class families, leading to significant annual savings of **₹15,000-₹18,000** on electricity bills and a typical payback period of **5-6 years**.

Accelerating Renewable Energy Goals

Supporting India's commitment to achieve **500 GW of non-fossil fuel energy capacity by 2030** (with significant progress already made) and reach **Net Zero emissions by 2070**.

SOLUTION DEVELOPMENT

The RTSE Platform Solution

Rooftop Solar Explorer

- BSES launched an innovative geospatial platform integrating high-resolution imagery, Digital Surface Models, and 3D point cloud data to assess rooftop solar potential across its Licensee areas
- The platform combines consumer electricity profiles, energy tariffs, and grid data to deliver accurate assessments at building, city, and regional levels.

Platform Impact

10,000+ users evaluated their rooftop potential in six months

Target: Solarize 1 lakh households by 2026-27

Coverage: Complete licensee area mapping with

with shadow-free analysis

Technical Approach –Part-1

GIS Based Mapping- User Consumer GPS, Shadow Free Analysis, Optimal System sizing

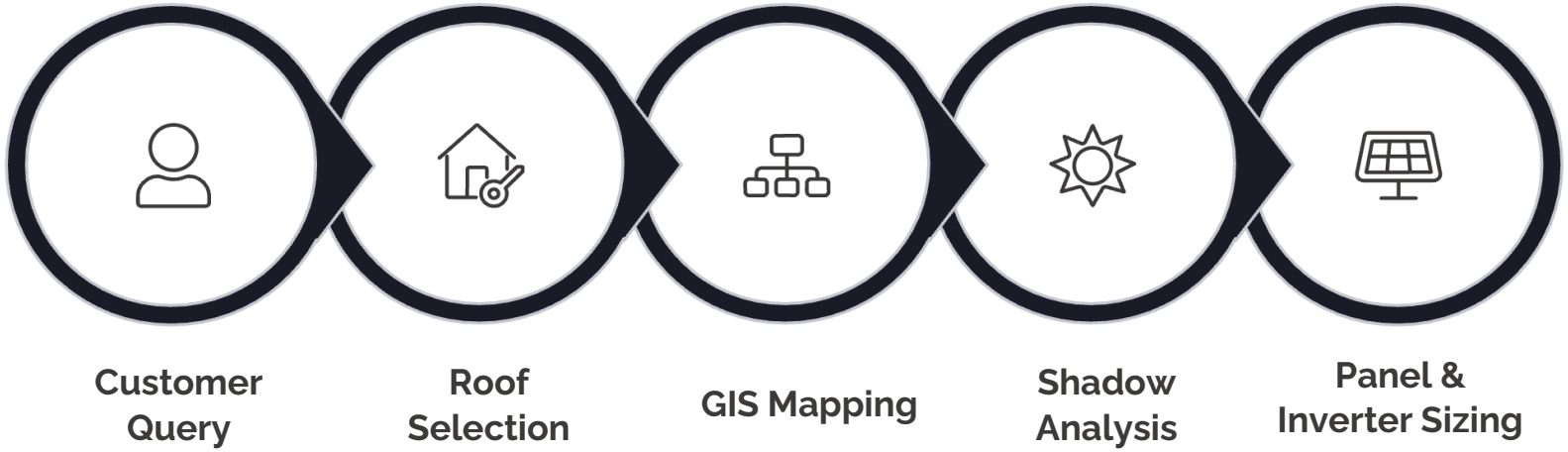
Data Inputs- Satellite Imagery**,** DSM, Consumption Profile, tariffs.

Financial Analysis- Part 2

Financial Modelling- Capex ,Opex, Payback(5-6 Years)

Key Parameters -60-65k Per Kw**,** Subsidy 30%-60% depending on size,25 Years Projection for feasibility

Design ,Installation & Proposal Workflow



Process : select consumer → analyze 1-year consumption → select roof → GIS-based design with obstacle insertion → shadow analysis → select panels/inverters → price & finance → generate proposal.

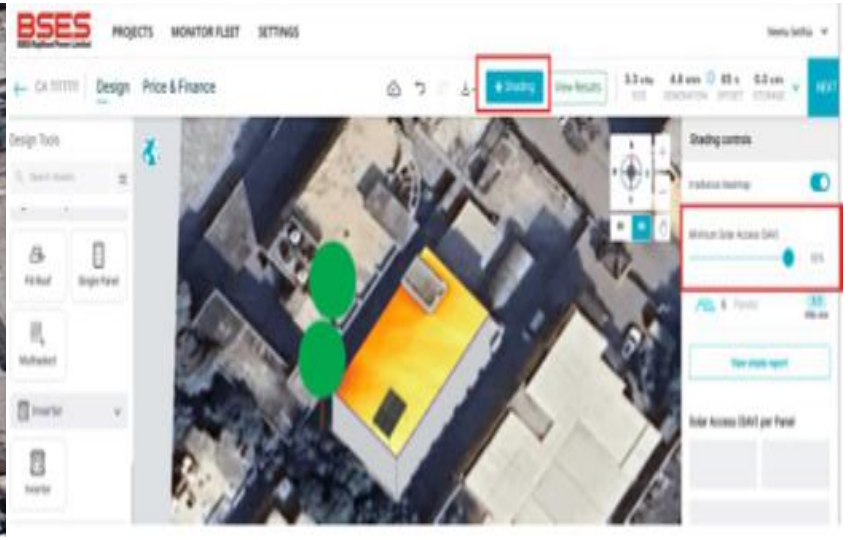
Design Support Link : https://support.enact-systems.com/portal/en/kb/articles/step-by-step-guide-for-designing-pricing-a-solar-pv-system-in-google-maps-4-6-2024

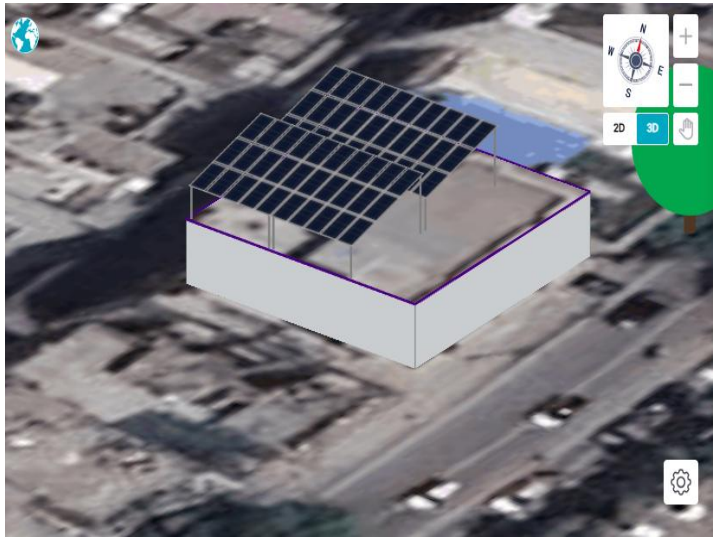
Sample Proposal :

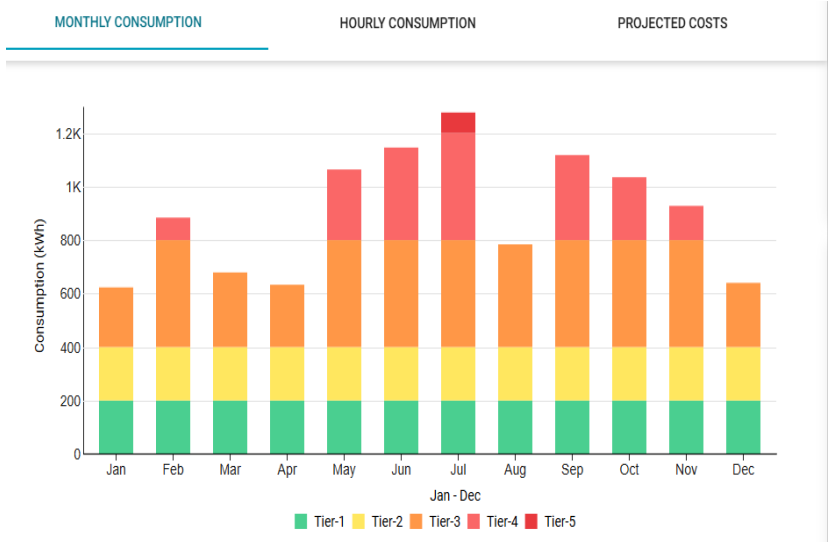
sample Customer Proposal

Roof Top solar Awareness Cum registration Camp held in Kakrola Village, Dwarka

CASE STUDY-CONSUMER IMPLEMENTATION



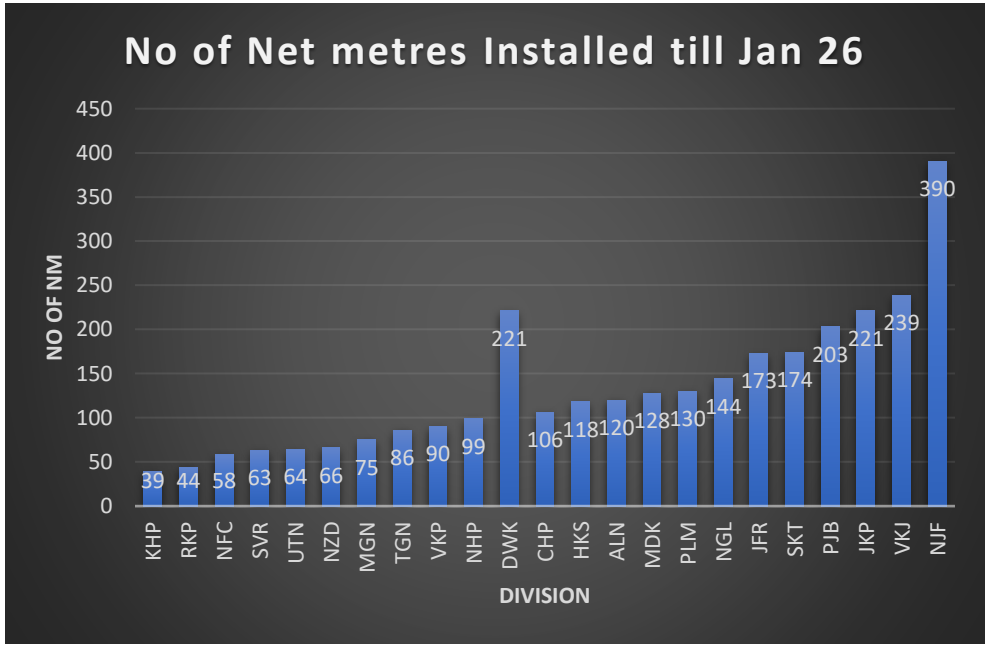




Implementation Results

- Annual consumption analysis revealed **peak demand during summer months (June-August)**, guiding optimal system sizing.
- GIS-based roof selection identified shadow-free areas, enabling precise panel placement for maximum efficiency.
- Financial modeling confirmed viability with substantial cost savings on electricity bills

Outcomes, Challenges ,Recommendation & Conclusion



Jan'26

No of NM Installed in Dwk is 221 till Jan'26

No of Proposals generated in Dwk is 410 till Jan'26

Outcomes

Significant bill savings; institutions achieved ~60% daytime demand from solar; carbon footprint reduced

Challenges

Procurement delays, limited skilled installers, maintenance under harsh weather, and financing constraints

Recommendations

Streamline net metering, offer soft loans, DISCOM incentives, capacity building, and continued RTSE deployment

Conclusion & Next step

Solar transition is vital for Delhi's energy security and sustainability. With RTSE, supportive policy, and finance, Delhi can scale rooftop solar and set a national example

- Scale RTSE across more cities
- Policy simplification & financing support
- Further research and monitoring for continual improvement

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)